

MAT 1272 – Statistics Exam 1 Practice Test

Exam 1: September 28, 2026

Exam information. The exam will be approximately **100 minutes**. Please be on time. A **TI-84 calculator is required**. Cell phones and other electronic devices are **not permitted during the exam**.

Important calculator requirement. When a question requires the TI-84, you must **write the calculator command(s) you used, in the order in which you used them** to obtain your answer. A numerical answer by itself may not receive full credit. For example, acceptable calculator work may include commands such as `sum(L1)`, $L2 = L1^2$, `sum(L2)`, or `1-Var Stats L1`, depending on the question.

Instructions: Show your work and clearly label your answers. This practice test reviews the material for Exam 1. The Empirical Rule is **not included**.

1. Variables and Types of Data

A college records the following information for each student: major, number of courses taken, weekly study time, and height.

- (a) Classify each variable as **qualitative/categorical** or **quantitative**.
 - (b) For each quantitative variable, classify it as **discrete** or **continuous**.
 - (c) Explain the difference between a discrete and a continuous variable.
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2. Population, Sample, and Variable

A professor wants to estimate the number of hours per week that City Tech students spend studying mathematics. She surveys 45 City Tech students and records their weekly mathematics study time.

- (a) Identify the population.
 - (b) Identify the sample.
 - (c) Identify the variable.
 - (d) Is the variable qualitative or quantitative?
 - (e) If quantitative, is it discrete or continuous?
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3. Organizing Qualitative Data

Twenty students were asked how they usually travel to campus.

Transportation	Frequency
Subway	8
Bus	5
Car	4
Walking	3

- (a) Find the relative frequency of each category.

- (b) Express each relative frequency as a percentage.
 - (c) Which category would have the largest bar in a bar graph?
 - (d) What percentage of students do *not* travel by subway?
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4. Stem-and-Leaf Display

Consider the test scores

52, 68, 71, 75, 64, 87, 83, 79, 92, 95, 61, 72, 86, 77, 98.

- (a) Construct a ranked stem-and-leaf display.
- (b) What is the smallest score?
- (c) What is the largest score?
- (d) How many scores are in the 70s?

5. Summation Notation and TI-84 Lists

Let

$$L1 = \{2, 3, 5, 7\}.$$

- (a) Find $\sum x$.
- (b) Find $(\sum x)^2$.
- (c) Find $\sum x^2$.
- (d) Explain why $(\sum x)^2$ and $\sum x^2$ are different.
- (e) Write the TI-84 command sequence you would use to calculate $\sum x$ from data stored in $L1$.
- (f) Describe how to create $L2 = L1^2$, and then write the command used to calculate $\sum x^2$.

6. Mean, Median, and Mode

For the data

$$12, 15, 15, 17, 20, 21, 26,$$

find:

- (a) the mean,
- (b) the median,
- (c) the mode.
- (d) Use the TI-84 to check the mean and median. Write the calculator command(s) you used, in the order used.

7. Effect of an Outlier

The selling prices of eight homes are

$$245,670, 176,200, 360,280, 272,440, \\ 450,394, 310,160, 393,610, 3,874,480.$$

- (a) Identify the obvious outlier.
- (b) Use your TI-84 to find the mean and median of all eight prices.
- (c) Write the TI-84 calculator command(s) you used to obtain these values, in the order used.
- (d) Which measure, mean or median, better represents the center of these prices? Explain.
- (e) Is the mean resistant or nonresistant to outliers?

8. Range and Standard Deviation

The earnings (in thousands of dollars) before taxes for **all six employees** of a small company are

$$88.50, 108.40, 65.50, 52.50, 79.80, 54.60.$$

- (a) Find the range.
- (b) Are these data a sample or a population? Explain.

- (c) Use the TI-84 to find the appropriate standard deviation.
 - (d) Write the calculator command(s) you used, in the order used.
 - (e) Should you report S_x or σ_x ? Explain.
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9. Sample or Population Standard Deviation?

A researcher randomly selects 10 City Tech students and records their commute times. After entering the data into a TI-84 and running 1-Var Stats, the calculator displays

$$S_x = 14.6 \quad \text{and} \quad \sigma_x = 13.9.$$

- (a) Is the data set a sample or a population?
 - (b) Which standard deviation should be reported?
 - (c) Explain the difference between S_x and σ_x .
 - (d) What does a larger standard deviation tell us about the spread of a data set?
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10. Putting It Together: One-Variable Statistics

A sample of eight students reports the following number of minutes spent commuting to campus:

25, 40, 32, 55, 28, 35, 42, 63.

- (a) Enter the data into $L1$.
- (b) Use the TI-84 to find the mean and median.
- (c) Find the range.
- (d) Find the appropriate standard deviation.
- (e) Write the TI-84 calculator command(s) you used to obtain your one-variable statistics, in the order used.
- (f) Explain why S_x , rather than σ_x , is appropriate for this problem.

TI-84 Review

Be comfortable with entering and editing lists, `sum(L1)`, creating $L2 = L1^2$, `sum(L2)`, and `1-Var Stats L1`.

On the exam, when calculator work is requested, **show the calculator command(s) you used and the order in which you used them.**